

1. The Data Imbalance Assumption

- **Initial Assumption:** Training models directly on a raw, real-world Amazon text scrape will yield a system that understands all star-ratings equally well.
- **The Reality:** Real data is heavily skewed toward 4 and 5-star reviews. Models exploit this imbalance by predicting the majority classes, triggering an "Accuracy Paradox" where overall accuracy looks high but minority classes (2 and 3 stars) score near-zero.
- **How it was Overcome:** The raw 50k data was filtered down into a strictly balanced 40,000-sample sub-corpus containing an exact 20% stratified allocation per rating class.

2. The Deep Learning Superiority Assumption

- **Initial Assumption:** Advanced, context-aware pre-trained deep learning features (like BERT) will automatically outperform simple word-counting techniques (like TF-IDF).
- **The Reality:** Out-of-the-box static BERT embeddings encode general semantic topics rather than sentiment polarity. The tuned TF-IDF pipeline actually outperformed BERT (47.16% vs. 42.86% accuracy) because it targeted explicit sentiment keywords.
- **How it was Overcome:** Grid searches were run on both feature extractors to isolate their specific sweet spots, proving that simpler, localized vocabulary control can beat static deep learning embeddings on text-ranking tasks.

3. The Dataset Scaling Assumption

- **Initial Assumption:** Model accuracy maps linearly to dataset size; reducing the training data will result in a smooth, gradual decline in performance.
- **The Reality:** Performance follows a sharp, logarithmic dependency. The system handles data reduction steadily down to 5,000 samples, but encounters a catastrophic "data starvation cliff" below 2,000 samples.
- **How it was Overcome:** Comprehensive macro and micro learning curves were plotted to pinpoint the absolute minimum threshold (2,000 instances) required before the sparse input matrix drops into statistical uselessness (24.16% accuracy at 500 samples).

4. The Model Complexity Assumption

- **Initial Assumption:** A bigger, deeper neural network architecture with multiple dense hidden layers is universally superior for parsing text vectors.
- **The Reality:** High-capacity neural networks are extremely sensitive to vector sparsity. Feeding a massive 5,000-dimensional sparse TF-IDF matrix into complex hidden structures leads to severe overfitting.
- **How it was Overcome:** The grid search proved that a simpler, shallow network (hidden_layer_sizes=(100,)) paired with heavy weight penalties (alpha=0.1) was required to stabilize predictions.

5. The Cleaning/Stop-Words Assumption

- **Initial Assumption:** Removing common "stop-words" (like *the*, *and*, *at*) from a text corpus is a mandatory preprocessing step that always boosts model performance.
- **The Reality:** While stripping stop-words helps sparse tokenizers focus on core keywords, doing so completely destroys the continuous semantic string structure required by transformer embeddings like BERT.
- **How it was Overcome:** An explicit ablation study was woven into the execution scripts to test pipelines both with and without stop-words, tracking exactly how and when lexical cleaning choices impact model variants.